

Arnav Samal

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Education

Present Nov 2022	National Institute of Technology (NIT) Rourkela Bachelor of Technology, Computer Science & Engineering CGPA: 9.21/10 <i>Coursework: Probability & Statistics, Computer Vision, Natural Language Processing, Machine Learning, Deep Learning</i>	Rourkela, India
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Research Experience

Present Jan 2026	IISc Bangalore Visual Computing Lab, Dept. of CDS  <i>Research Intern Advisor: Dr. Anirban Chakraborty</i> > Driving experimental validation in Federated Learning to analyze how model overparameterization mitigates the effects of data heterogeneity across large-scale models > Investigating the capacity of LLMs to extrapolate sequential 2D sketches into complex 3D surfaces and curves.	Bengaluru, India
Dec 2025 July 2025	NIT Rourkela Dept. of CS <i>Undergraduate Researcher Advisor: Dr. Prasenjit Dey</i> > Developed architectural modifications to diffusion models for sketch-to-face synthesis to enhance geometric faithfulness and feature mapping. > Integrated explainability mechanisms to ensure interpretable generation of facial features from sketches.	Rourkela, India
Dec 2025 May 2025	IIT Hyderabad Lab 1055 (AI & Vision-Language Group), Dept. of CS <i>Research Intern Advisor: Prof. Vineeth N Balasubramanian, in collaboration with Microsoft Research</i> > Investigated concept quality in CBMs by analyzing concept locality to diagnose concept leakage; processing 2TB+ medical imaging data and applying locality-to-concept specific metrics > Results show models attend 45% more to spurious regions than diagnostically relevant features	Hyderabad, India
Aug 2024 May 2024	IIT Hyderabad Data-driven Intelligence & Learning Lab, Dept. of AI  <i>Research Intern Advisor: Dr. Konda Reddy Mopuri</i> > Conducted comparative analysis of post-hoc explainability techniques in Vision Transformers, including attention rollout and token pruning methods. > Implemented and evaluated token pruning strategies for computational efficiency while maintaining classification performance.	Hyderabad, India

Selected Projects

SketchWarp Paper / Source Code	Mar'25 - May'25
> Developed a self-supervised learning framework in PyTorch for dense photo-to-sketch correspondences, enabling automatic image-to-sketch warping. > Designed and implemented training and evaluation pipelines inspired by the “<i>Learning Dense Correspondences between Photos and Sketches</i>” paper.	
Capsule Vision Challenge 2024 Team Seq2Cure Preprint / Competition Website / Source Code	Sep'24 - Nov'24
> Developed a fine-tuned, multi-model ensemble combining CNN and Transformer architectures, leveraging techniques like weighted random sampling and focal loss to address significant class imbalance. > The proposed method secured the 5th position in the international challenge, resulting in co-authorship in the official challenge summary paper.	
NeurIPS Ariel Data Challenge 2024 Team Markov's Chain Competition Website / Source Code	Aug'24 - Oct'24
> Developed a pipeline for exoplanet spectral prediction using calibrated multi-sensor time-series data, implementing spatial-temporal aggregation, phase detection via gradient analysis, and Nelder-Mead optimization. > Ranked in the top 20% in the NeurIPS-Ariel Challenge, with an evaluation score of 0.5704.	

Technical Skills

- > **Programming Languages:** Python, C++, C, SQL
- > **Libraries & Frameworks:** PyTorch, JAX, HuggingFace (Transformers, PEFT/LoRA, Datasets), NumPy, SciPy, scikit-learn, OpenCV, timm, Matplotlib, W&B
- > **Tools & Platforms:** Git, GitHub, uv, Conda, Docker, Jupyter, LaTeX, Microsoft SQL Server

Honours and Awards

Departmental Rank #5 [🏆] Ranked 5th in the Computer Science & Engineering department at NIT Rourkela based on cumulative academic performance.

ACM-IKDD Uplink 2025 Program [🏆] Selected for the competitive CS mentorship program (acceptance rate ~2%).

Research in Intelligence & Security Challenges (RISC) 2025 Program [🏆] Selected for the prestigious program at the University of Maryland.

Worldwide Rank 5 | Capsule Vision Challenge 2024 [🏆] Top-performing team (Team Seq2Cure) in the Capsule Vision Challenge organized by CVIP 2024.

2nd Place | HackFest 2024 Secured 2nd place at the [hackathon](#) organized by ML4E for undergraduates.

Kaggle Expert [🏆] Earned expert status for contributions in Datasets & Notebooks.

Leadership & Extra-Curricular Activities

ML4E Society, NIT Rourkela *Core Member* Aug'23 – Present

- > Helped establish and grow a student-led machine learning society; conducted sessions, mentored juniors, and contributed to research discussions.

Inquizzitive, NIT Rourkela *Quizzer* Sept'23 – Present

- > Active member of the quizzing society; participated in quizzes and contributed to discussions on trivia and strategy.

Rotaract Club, NIT Rourkela *Member* Oct'23 – Mar'24

- > Contributed to social initiatives, helping organize and participate in community outreach and development events.

Innovision 2K23, NIT Rourkela *Event Coordinator, Organizer* Sept'23 – Nov'23

- > Co-organized events for the annual technical fest, coordinating logistics and team operations.